29th Sept. Lecture-12

Recap:

1) Sets → Conventional way
   → Predicate based

2) Standard sets:
   $\phi, \mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$

3) $a \in A$ & $a \notin A$, A set

- $\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\} := \mathbb{Q}$.
  - It is read as the set of elements of the type $\frac{a}{b}$, for every $a, b \in \mathbb{Z}$, where $b \neq 0$.

4.1 Comparing sets

Let S and T be sets. We define some common terminology.

a) The notation $S \subseteq T$ means $\forall x \in S \ (x \in T)$. In this case,

Quantifier Domain Predicate

S is called a subset of T.

Equivalently, $\forall x \in S \ (x \in T)$ can be written as

$\forall x \in U (x \in S \Rightarrow x \in T)$

where the domain of x is every element under consideration. formally, it is captured by the universal set.

If $A \subseteq B$ then B is called a super set of A.

Definition [Universal set]: A set that contains every element

under consideration. This set is denoted as U.

- The notation $S \nsubseteq T$ means that $\exists x \in S [x \notin T]$. In this case, we say that S is not a subset of T.

E.g.: $\mathbb{N} \subseteq \mathbb{Z}$ but $\mathbb{Z} \nsubseteq \mathbb{N}$

b) The notation $S \subsetneq T$ means that $(S \subseteq T) \wedge [\exists y \in T (y \notin S)]$. In this case, S is called a proper subset of T.

E.g.: $\mathbb{N} \subsetneq \mathbb{Z}$.

Definition [Equality of sets]: Two sets S & T are equal, denoted

$S = T$ if and only if $S \subseteq T$ and $T \subseteq S$.

Remark! Often, equality of sets is misunderstood.

Example: Let $S := \{2x : x \in \mathbb{Z}\}$

and $T := \{x \in \mathbb{Z} : \exists y \in \mathbb{Z}$ such that $x = 2y\}$

Claim 4.1.1 : $S = T$

Proof: a) To show $S \subseteq T$:

We want to prove

$\forall m \in S \; (m \in T)$ is true.

Pick $m \in S$ arbitrarily. Then, by definition of S, m looks

as $m = 2x$, where $x \in \mathbb{Z}$. By the definition of T, $m \in T$. Hence, $S \subseteq T$.

b) To show that $\boxed{T \subseteq S}$ we want to prove $\forall m \in T\ (m \in S)$. Let us pick an $m \in T$ arbitrarily. Then, there exists a $y \in T$ s.t. $m = 2y$. Thus, by the definition of S, $m \in S$. Hence $T \subseteq S$.

As $S \subseteq T$ and $T \subseteq S$, we get $S = T$.

Homework: Let $S = \{n \in \mathbb{Z} : n + 5 \text{ is odd}\}$

and $T = \{n \in \mathbb{Z} : n \text{ is even}\}$. Prove that $S = T$.

→ [Transitivity of $\subseteq$]

Observation 4.1.2: Let A, B, C be arbitrary sets such that $A \subseteq B$, $B \subseteq C$. Then, $A \subseteq C$.

Proof: As $A \subseteq B$, $\forall x \in A (x \in B)$ is true. Similarly, as $B \subseteq C$, $\forall y \in B (y \in C)$ is true. Thus, we get $\forall x \in A (x \in C)$ is also true. Hence, $A \subseteq C$. ∎

Claim 4.1.3: for every set A, $\phi \subseteq A$.

Proof: Let A be an arbitrary set. We want to show $\phi \subseteq A$, which is equivalent to

$\forall x \in U (x \in \phi \Rightarrow x \in A)$, where U is the universal set. As ϕ does not contain any element, $x \in \phi$ is false for every $x \in U$. Thus, from the truth table of $\Rightarrow$, we get that $\forall x \in U (x \in \phi \Rightarrow x \in A)$ is true. ☐

4.2 Combining sets: Operation on sets.

Now, we discuss how to create new sets from existing sets. This is similar to creating new propositions using existing. We will see that the set operations given below are tightly linked with logical operations. Let A, B be sets.

1) Union: The union of A & B denoted $A \cup B$ is the set defined as follows

$A \cup B := \{x : x \in A \vee x \in B\}$

operator & not the universal set.

definition

$\{1,2,a\} = \{1,2,\top,a\} = \{1,2,\top,a\}$

$A = \{x : x \in A \text{ or } x \in B\}$

$A = \{1,2\}$
$B = \{1,a\}$

2) **Intersection:** The intersection of A & B, denoted $A \cap B$ is defined as follows

$$A \cap B := \{x : x \in A \wedge x \in B\}$$

3) **Set-difference:** The difference of B and A, denoted $B - A$ (also denoted $B \setminus A$) is the set defined as follows

$$B - A := \{x : x \in B \wedge \underbrace{x \notin A}_{\neg(x \in B)}\}.$$

Similarly,

$$A - B := \{x : x \in A \wedge x \notin B\}.$$

4) **Complement:** Let U be the universal set. Then, the complement of A is $U-A$. It is also denoted as $\overline{A}$ and defined as

In 4, $\overline{A} := \{x : x \in U \wedge x \notin A\}$.

Remark: We can U with any B s.t $A \subseteq B$.

5) **Cartesian product:** The cartesian product of A & B, denoted $A \times B$ is defined as follows

$$A \times B := \{(a,b) : a \in A \wedge b \in B\},$$

where (a,b) is the ordered pair containing a as the first element & b as the second element.

Definition [Equality of pairs]:

Let (a, b), (c, d) be two pairs. We say that they are equal denoted $(a, b) = (c, d)$ if and only if $a = c \wedge b = d$.

- In general $(a, b) \neq (b, a)$.

$(1,2) \neq (2,1)$

$\{1,2\} = \{2,1\}$

Example: Let $A = \{1, 2\}$, $B = \{a, b\}$. Then,

a) $A \cup B = \{1, 2, a, b\}$

b) $A \cap B = \emptyset$

c) $A - B = \{1, 2\}$

d) $B - A = \{a, b\}$

e) $A \times B = \{(1,a),(1,b),(2,a),(2,b)\}$

f) $B \times A = \{(a,1),(a,2),(b,1),(b,2)\}$

$\neg(A=B) \Leftrightarrow \neg(A \subseteq B \wedge B \subseteq A)$

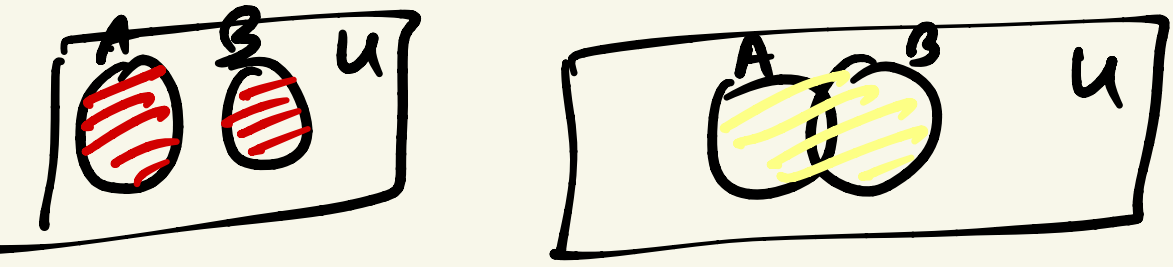
Note that here $\boxed{A \times B \neq B \times A}$.

Remark: We say that two sets A, B are <u>not equal</u>, denoted $A \neq B$ if either $A \nsubseteq B$ or $B \nsubseteq A$.

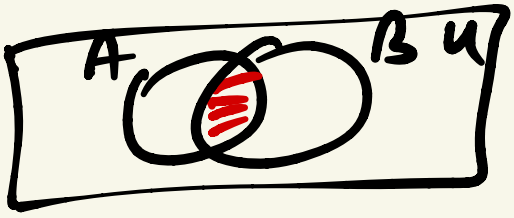
- **Venn diagrams:** A venn diagram is a pictorial representation of various operations on sets.
  - Here sets are represented as circles & the universal

Set as a square.

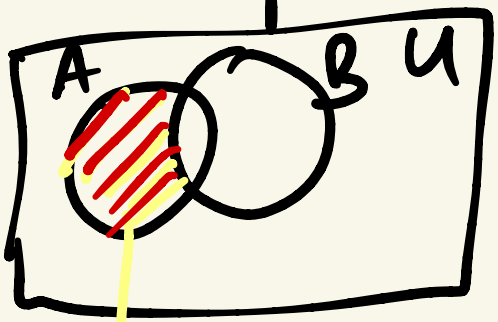
a) Union : $A \cup B$

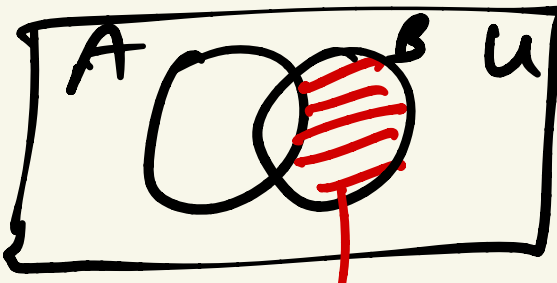


b) Intersection: $A \cap B$



c) Complement: $A - B$ & $B - A$



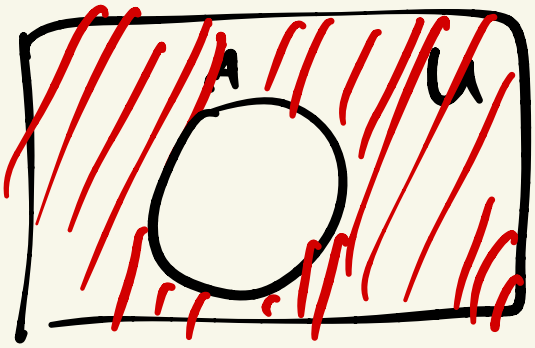


$A - B$

$B - A$.

d) Complement: $\bar{A}$

Remark: Do not use Venn diagrams to prove anything about sets.



Theorem 4.2.1: Let A, B, C be subsets of the universal set U.
↳ arbitrary.

Then,

a) $A \cup \phi = A$ and $A \cap \phi = \phi$

b) $A \cap B \subseteq A$ and $A \subseteq A \cup B$

c) $A \cup \bar{A} = U$ and $A \cap \bar{A} = \phi$

d) $\bar{\bar{A}} = A$

e) [Idempotence]: $A \cup A = A$ and $A \cap A = A$

f) [Commutativity]: $A \cup B = B \cup A$ and $A \cap B = B \cap A$.

g) [Associativity]: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.

h) (Distributivity of $\cup$ & $\cap$):

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

& $$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

i) (De Morgan's laws):

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \quad \text{and}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}.$$

Home work: Prove these.

We are dealing with equality of sets.

Mid-sem answer key

Que 1.1.1 : follow the priority order of logical operators. ==

b) $p \Rightarrow \neg q \vee r \wedge \neg p \Leftrightarrow r$

1.1.2 : a) There exists a student at IIIT D who knows singing or dancing.

Answer : x is a variable,

domain (x) = all students of IIIT D

P(x) := x knows singing

$Q(x)$:= x knows dancing

$\exists x\,(P(x) \vee Q(x))$

- $\neg \exists x\,(P(x) \vee Q(x))$
  $\equiv \forall x\, \neg(P(x) \vee Q(x))$ [De Morgan's law of quantifiers]

  $\equiv \forall x\,(\neg P(x) \wedge \neg Q(x))$ [De Morgan's law]

Every student of IIITD doesn't know singing & doesn't know dancing.

b) Every Indian citizen enjoys fundamental rights & beats

fundamental duties.

x : Domain (x) = all Indian citizens.

$F(x)$: x enjoys fundamental rights

$D(x)$: x bears fundamental duties.

$\forall x \, (F(x) \wedge D(x))$.

Que 1.1.3 x, y variables

$$\exists x \forall y \, P(x,y) \overset{?}{\equiv} \forall y \exists x \, P(x,y)$$

for every predicate $P(x,y)$ and for every domain of x & y. False

Counterexample:

$P(x,y) = x - y > 0$

x, y are integers.

Que 1.1.4.: Converse, inverse & contrapositive & which of these are equivalent?

Que 1.2.1: Given propositional variables $p_1, \ldots, p_n$. Does there exist a propositional formula u in $p_1, \ldots, p_n$ s.t. u is a tautology?

Yes. $\underbrace{p_1 \vee \neg p_1}_{T} \vee \underbrace{(p_2 \cdots \vee p_n)}_{W}$

$T \vee W \equiv T$

Clearly, u is a tautology.

Que 1.2.2 Convert $(p \oplus q) \in \pi$ to equivalent conjunctive formula using only the algebra of propositional logic.

CNF was not required.

Que 1.2.3: Are the following two formulas equivalent?

$$u_1 := (p_1 \vee \neg p_2) \wedge (p_2 \vee \neg p_3) \wedge \cdots \wedge (p_{n-1} \vee \neg p_n) \wedge (p_n \vee \neg p_1)$$

$$u_2 := ((p_1 \wedge p_2 \wedge \cdots \wedge p_n) \oplus (\neg p_1 \wedge \neg p_2 \wedge \cdots \wedge \neg p_n))$$

Ques 1.3.1: a) $a, b, c, d \in \mathbb{Z}$ s.t.

$$a + b + c = d.$$

Prove that d is odd if & only if either exactly one of a, b, c is odd or each of a, b, c is odd.

Ques 1.3.2. a) False.

b) $P(x)$: $x+5 \geq 10$

domain of x: all odd numbers.

$\neg \exists x\, P(x)$: for every even integer x, $x+5 < 10$.

False.

c) false.

One 1.4 ✓

One 1.5.1. Given that there are finitely many primes $p_1, \ldots, p_n$. Does there exist a natural

number b, $b \neq 1$ s.t.

$p_1 \cdots p_n + b$ is a prime.

Homework: under the same condition, is

$p_1 \cdots p_{n-1} + p_n$

a prime number?

Answer: false.

for every $b \neq 1$,

$p_{n+1} := p_1 \cdots p_n + b$ is not prime.

Case 1: $b = p_i$

$p_{n+1} = p_i (\neq 0)$

Composite

Case 2: b is composite

By fact A.1, $\exists$ p_i s.t. $p_i < b$ & $p_i \mid b$.

$p_{n+1} = p_i \square \neq 0$.

$\square$

Que 1.6

$A = \{ 2^{\frac{k}{2}} \mid \exists\, m \in \mathbb{N} \text{ s.t. } k = m^2 \}$.

Does A have finite irrational numbers?

1) Prove that there are infinitely many odd numbers!

Suppose there are n odd numbers $o_1, \ldots, o_n$.

Suppose $o_1 < o_2 < \cdots < o_n$.

$o_n + 2$ is
- odd
- greater than every o_i.

2) there are infinitely many square of odd numbers.

Suppose there are n such numbers

$$o_1^2 < o_2^2 < \dots < o_n^2$$

$o_n^2 + 2$ X

$(o_n + 2)^2$

3) $2^{\frac{k^2}{2}}$ is irrational for every odd k.

suppose not. $a, b \in \mathbb{Z}$

$$2^{\frac{k^2}{2}} = \frac{a}{b}, \quad b \neq 0$$

$$2^{k^2} \cdot b^2 = a^2.$$

$\Rightarrow$ a^2 is even

$\Rightarrow$ a is even

$\Rightarrow$ b^2 is even

$\Rightarrow$ b is even

requires care

- Let a be rational, b be irrational.
$a \cdot b$ is irrational.

$$a \cdot b = \frac{m}{n}$$

we know $a = \frac{p}{q}$

$$\frac{p}{q} \cdot b = \frac{m}{n}$$

$$b = \frac{\square}{\square} \times$$

5) There are infinitely many irrationals in $\{2^{\frac{k^2}{2}} \mid k$ is odd natural numbe

Suppose not. There are n irrationals $2^{\frac{k_1^2}{2}}, \ldots, 2^{\frac{k_n^2}{2}}$ for some n.

Assume $k_1 < k_2 < \ldots < k_n$.

$2^{\frac{(k_n+2)^2}{2}}$ is not in the list & it is irrational.

1st October Lecture-13

Recap:

a) Sets → Conventional representation
→ Predicate based

b) subsets, proper subsets

c) Popular sets: ϕ, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$

d) Transitivity of $\subseteq$

*** e) Equality of sets: $A = B$ if and only if $A \subseteq B$ & $B \subseteq A$.

f) Operations on sets:

i) Union ($A \cup B$)

ii) Intersection ($A \cap B$)

iii) Set-difference ($A - B$)

iv) Complement ($\bar{A}$)

v) Cartesian Product ($A \times B$)

g) ϕ is a subset of every set.

h) Universal set & Venn diagrams

Definition [Disjoint sets]: Two sets A & B are said to be disjoint if and only if $A \cap B = \phi$.

Example: Let E & O be the sets of even & odd integers. Then, E & O are disjoint sets.

- A collection of sets $A_1, \ldots, A_n$ are said to be <u>disjoint</u> if and only if for every $i, j \in \{1, \ldots, n\}$ $i \neq j$, $A_i \cap A_j = \phi$.

Definition [Power set]: Let A be a set. Then, the power set of A, denoted $\text{pow}(A)$, is the set of all subsets of A. In other words, $B \in \text{pow}(A)$ if and only if $B \subseteq A$.

Example: Let $A = \{1, 2\}$. Then,

$$\text{pow}(A) = \{\phi, \{1\}, \{2\}, \{1,2\}\}.$$

Note that $1 \notin \text{pow}(A)$ but $\{1\} \in \text{pow}(A)$. There is a difference between 1 & $\{1\}$ as the former one is an element while the latter one is a set.

H.W.
Let $A = \{\{1\},\{2\}\}$
What is pow(A)?

ii) Let $A = \phi$.

$pow(A) = \{\phi\}$ (Homework Is this correct?)

Fact: Let A be a set containing n elements. Then pow(A) contains 2^n elements.
We will prove this fact in Combinatorics.

Theorem 4.1.1 [Properties of union, intersection & set-difference]: Let A, B, C be [non-empty] subsets of the universal set U. Then,

1) $A \cup \phi = A$ & $A \cap \phi = \phi$.

ϕ & ϕ : Homework

Proof of 1). We want to show that $A \cup \phi = A$. To do this, we have to show that $A \cup \phi \subseteq A$ & $A \subseteq A \cup \phi$. Let $a \in A \cup \phi$ be an arbitrary element. It means $a \in A \vee a \in \phi$. As $a \in \phi$ is false, the identity rule for OR implies

$$a \in A \vee \overbrace{a \in \phi}^{F} \equiv a \in A$$

Hence, $a \in A$ & as a was arbitrary, we get $A \cup \phi \subseteq A$. Now, let $a \in A$ be an arbitrary element. Then, by the identity rule for OR, we have

$a \in A \equiv a \in A \vee F$.

As ϕ is the empty set, we have $\neg(\forall x \in U)$, $x \in \phi \equiv F$. Hence,

$a \in A \equiv a \in A \vee a \in \phi$.

Thus, $a \in A \cup \phi$. As a was arbitrary, we get $A \subseteq A \cup \phi$.

Similarly, we can show that $A \cap \phi = \phi$ (Homework).

2) $A \cap B \subseteq A$ & $A \subseteq A \cup B$.

Proof. We want to show that $A \cap B \subseteq A$. Let $a \in A \cap B$ be an arbitrary element. Then, $a \in A \wedge a \in B$ is true. It

follows from the truth table of AND that both $a \in A$ & $a \in B$ are true. As a was arbitrary, we get $A \cap B \subseteq A$. Similarly, using the truth table of OR, we can show that $A \subseteq A \cup B$ (Homework).

3) $A \cup \bar{A} = U$ & $A \cap \bar{A} = \phi$.

Proof: In the first part, we want to show that $A \cup \bar{A} \subseteq U$ and $U \subseteq A \cup \bar{A}$. Here U is the universal set. Let $a \in A \cup \bar{A}$ be arbitrary.

Then, $a \in A \vee a \notin A$ is true.
This means $a \in A$ or $a \in U - A$
is true. In both cases, $a \in U$
as U is the universal set.
Since a was arbitrary, we
get $A \cup \bar{A} \subseteq U$.
Now, let $a \in U$ be an
arbitrary element. Since U
is the universal set, by
definition either $a \in A$ or
$a \notin A$. This implies
$a \in A \vee a \in \bar{A}$ is true.
Hence, $U \subseteq A \cup \bar{A}$.
Similarly, we can prove $A \cap \bar{A} = \phi$
(Homework).

iv) $\overline{\overline{A}} = A$.

Proof: It can be prove using the property $\neg\neg p \equiv p$ for every proposition p.

(Homework: Write it formally).

v) (Idempotence). $A \cup A = A$ and $A \cap A = A$.

Proof: To prove the first part, we have to show that $A \cup A \subseteq A$ and $A \subseteq A \cup A$.

Let $a \in A \cup A$ be arbitary.

Then $a \in A \vee a \in A$ is true.

By the idempotence of OR, we get

$a \in A \vee a \in A \equiv a \in A$.

As a was arbitrary, we get $A \cup A \subseteq A$. Similarly, we can show $A \subseteq A \cup A$.

$A \cap A = A$ can be shown using the idempotence of AND (Homework)

vi) (Commutativity). $A \cup B = B \cup A$ and $A \cap B = B \cap A$.

Homework: Prove it

vii) (Associativity). $A \cup (B \cup C) = (A \cup B) \cup C$ & $A \cap (B \cap C) = (A \cap B) \cap C$.

Homework: Prove it

viii) (Distributivity): $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof: To show the first part, we have to prove that

$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and

$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

Let $a \in A \cap (B \cup C)$ be arbitrary. This means $a \in A \wedge (a \in B \vee a \in C)$ is true. By the distributivity of AND over OR, we get

$$a \in A \wedge (a \in B \vee a \in C) \equiv$$
$$(a \in A \wedge a \in B) \vee (a \in A \wedge a \in C)$$

As a is arbitrary, this implies

$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Similarly, we show that

$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$.

In the same way, we prove

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(Homework).

ix) (De Morgan's laws): $\overline{A \cup B} = \bar{A} \cap \bar{B}$ and $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

Proof. To show the first part, we want to show $\overline{A \cup B} \subseteq \bar{A} \cap \bar{B}$ and $\bar{A} \cap \bar{B} \subseteq \overline{A \cup B}$.

Let $a \in \overline{A \cup B}$. This means $a \notin A \cup B$. Equivalently, it is not the case that $a \in A$ or $a \in B$. Let p be the proposition that $a \in A$ and q be the

be arbitrary

proposition that $a \Sigma B$. Then, $a \not\models (A \cup B)$ is essentially $\neg(p \vee q)$.

By the DeMorgan's law,

$$\neg(p \vee q) \equiv \neg p \wedge \neg q.$$

Hence $a \not\models A \wedge a \not\models B$. As a was chosen arbitrarily, we get $\overline{A \cup B} \subseteq \bar{A} \cap \bar{B}$.

Similarly, we can show $\bar{A} \cap \bar{B} \subseteq \overline{A \cup B}$.

In the same way, we can prove $\overline{A \cap B} = \bar{A} \cup \bar{B}$ (Homework).

Remark: This theorem shows how set theory is developed from propositional logic.

Theorem 4.1.2: Let A, B, C, D be arbitrary sets (may be finite). Then,

1) $A \times \phi = \phi \times A = \phi$.

Proof: We want to show $A \times \phi \subseteq \phi$ & $\phi \subseteq A \times \phi$. The latter is true as ϕ is a subset of every set. Suppose $A \times \phi \neq \phi$. Let $(a, b) \in A \times \phi$. By definition, $a \in A \wedge b \in \phi$ is true. As ϕ is the empty set, $b \in \phi$ is false.

which implies $a \in A \wedge b \in \phi \equiv F$.
Hence it is a contradiction &
$A \times \phi \subseteq \phi$.

Similarly, we can show $\phi \times A = \phi$.

2) $A \times (B \cup C) = (A \times B) \cup (A \times C)$
and $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

Proof: Let $(a, b) \in A \times (B \cup C)$ be arbitrary element. Then,

$$a \in A \wedge (b \in B \vee b \in C) \equiv T.$$

By the distributivity of $\wedge$ over $\vee$, we get

$a \in A \wedge (b \in B \vee b \in C) \equiv$

$(a \in A \wedge b \in B) \vee (a \in A \vee b \in C)$.

This implies $(a, b) \in (A \times B) \cup (A \times C)$.

Hence, $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$.

Similarly, we can prove that

$(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$.

In the same way, we can show $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

(Homework).

3) $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Proof: Homework.

4) $(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D)$.

In general, the equality of

sets in this case does not hold.

Proof: Let $(a,b) \in (A \times B) \cup (C \times D)$

Then, $((a,b) \in A \times B) \vee ((a,b) \in (C \times D))$

This is equivalent to [Homework]

$(a \in A \wedge b \in B) \vee (a \in C \wedge b \in D)$.

By using the truth tables of AND and OR, we get that [Homework]

$(a \in A \vee a \in C) \wedge (b \in B \wedge b \in D)$.

Thus, $(a,b) \in (A \cup C) \times (B \cup D)$. ✓

As (a,b) was arbitrary, we get that

$(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D)$

Now, we show that the equality

does not hold by giving a counterexample.

Take $A = \phi$ and $D = \phi$, and B, C to be non-empty sets.

Then, by the first part,

$(A \times B) = \phi = (C \times D)$, which implies $(A \times B) \cup (C \times D) = \phi$.

But, $(A \cup C) = C$ and $(B \cup D) = B$, which implies $(A \cup C) \times (B \cup D)$ is non-empty. Hence,

$$(A \cup C) \times (B \cup D) \not\subseteq (A \times B) \cup (C \times D).$$

$(A \cup C) = C$, $(B \cup D) = D$

$C \times D$

∎

6th October Lecture-14

Recap

a) Sets, subsets, power sets, universal sets, proper subsets, super sets.

b) Operations on sets: $\cup, \cap, -, \times$.

c) Properties of various set operations. - Everything is obtained from propositional logic.

4.3 Sequences

Sets provide a way to group a collection of elements, but there is no order between the elements.

Another way to group elements is in the form of a sequence, where elements follow some ordering.

Definition [Sequence]: A sequence is a collection of elements, where elements need not be distinct and they follow an ordering.

Always denoted using ( ).

for example, (1,2,1,2,3) is a sequence.

- {1,2,3} and {3,2,1} are the same sets, but (1,2,3) and (3,2,1) are not the same sequences as they have different ordering.
- Sequences are the elements of

the cartesian product of finitely many sets. for example,

$(1,2,3) \in \mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}$, $(1,2,\frac{1}{2}) \in \mathbb{Z}\times\mathbb{Z}\times\mathbb{Q}$

Definition [Equality of sequences]. We say that two sequences $(x_1,\cdots,x_n)$ and $(y_1,\cdots,y_m)$ are equal, denoted $(x_1,\cdots,x_n)=(y_1,\cdots,y_m)$ if and only if $n=m$ and for every $i\in\{1,\cdots,n\}$, $x_i=y_i$.

for example, $(1,2,1)\neq(1,1,2)$.

4.4 Functions.

This is one of the most important parts of set theory and almost

every area of mathematics depend heavily on functions.

Definition [Function]: Let X, Y be two non-empty sets. A function F from X to Y, denoted $F: X \to Y$, is defined as a subset $f \subseteq X \times Y$ such that for every $x \in X$, there is a unique $y \in Y$ such that $(x, y) \in F$.

- **Some important notations**: Let $f: X \to Y$ be a function.
  1) for $x \in X$, the unique $y \in Y$ such that $(x, y) \in F$ is denoted by $f(x)$. Here, x is called the pre-image of $f(x)$ and $f(x)$ is called the image of x.

2) X and Y are called the domain and co-domain of the function f, respectively, denoted Domain(F) and co-Domain(F).

3) The set $\{f(x) \in Y : x \in X\}$, denoted $f(X)$, is called the range of F, denoted Range(f).

Set

Domain co-Domain

$F : X \to Y$

$F(X)$ is the range.

Examples:

1) Let $A = \{1, 2\}$ and $B = \{a, b, c\}$.

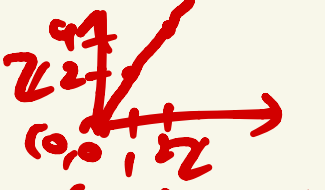
Let $f_1 = \{(1, a), (1, b), (2, c)\}$, ✗

$f_2 = \{(1, a), (2, b)\}$ and ✓

$f_3 = \{(1, a)\}$. Then, X

f_1 & f_3 are not functions & f_2 is a function.

Here Domain $(f_2) = \{1, 2\}$,
Range $(f_2) = \{a, b\}$.



$F = \{(0,0), (1,2), (2,4), \ldots\}$

2) Let $f : \mathbb{Z} \to \mathbb{Z}$ such that for every $x \in \mathbb{Z}$, $f(x) := 2x$.

Let us first argue that f is a function. We know by the closure properties of $\times$ that for every $x \in \mathbb{Z}$, $2x \in \mathbb{Z}$.

Now, we argue that for every $x \in \mathbb{Z}$, there is a unique y such that $f(x) = y$, i.e., $2x = y$

there exist $y, y' \in \mathbb{Z}$ such that $f(x) = y$ and $f(x) = y'$. Then, $2x = y = y'$. Hence, $f(x)$

is unique for every $x \in \mathbb{Z}$.

Hence, f is a function.

3) Let $f: \mathbb{R} \to \mathbb{R}$ (real numbers) such that for every $x \in \mathbb{R}$, $f(x) := e^x$.

H.W.: Assuming $e^x \in \mathbb{R}$, prove that f is a function.

Here, Range $(f) \subseteq \mathbb{R}^+$, where $\mathbb{R}^+$ is the set of all positive real numbers. ∎

Definition [Equality of functions]: Let A, B, C, D be non-empty sets and $f: A \to B$, $G: C \to D$ be functions. We say that f and G are equal, denoted $f = G$ if and only if the following properties hold.

a) $A = C$ and $\text{Range}(f) = \text{Range}(G)$ (i.e., equality of the domains of ranges of f & G),

b) for every $x \in A$, $f(x) = G(x)$.

If either of these properties do not hold then, $f \neq G$.

We continue with the examples of functions.

4. Let $f: \mathbb{R} \to \mathbb{R}$ and $G: \mathbb{R} \to \mathbb{R}$ such that for every $x \in \mathbb{R}$, $f(x) = \sqrt{x^2}$ and $G(x) = x$.

Que: Is $f = G$?

No. $F(-6) \neq G(-6)$.

5) **Identity function:** Let X be a non-empty set. Then, the identity function on X, denoted $I_X : X \to X$, is defined as: for every $x \in X$, $I_X(x) = x$.

6) **Polynomial function:** Let $n \in \mathbb{N}$, $a_0, a_1, \ldots, a_n \in \mathbb{R}$ such that $a_n \neq 0$. Then, a function $f: \mathbb{R} \to \mathbb{R}$ is said to be a polynomial of degree n with coefficients $a_0, \ldots, a_n$ if and only if for every $b \in \mathbb{R}$,

$$f(b) = a_0 + a_1 b + a_2 b^2 + \cdots + a_n b^n.$$

We first prove that f is a function. We know that the sum and product

of any two real numbers is also a real number. Then, for every $b \in \mathbb{R}$, $f(b) \in \mathbb{R}$. Now we want to show that for every $b \in \mathbb{R}$, $f(b)$ is unique. Suppose $y, y' \in \mathbb{R}$ satisfy $f(b) = y = y'$. Then, we get

$$y - y' = (a_n - a_n) b^n + \cdots + (a_1 - a_1) b + (a_0 - a_0)$$

$$= 0.$$

Hence $y = y'$. ✓

Thus, f is a function. Polynomial functions are often called polynomials and are extensively studied in mathematics & computer science.

9) Let $f: \mathbb{R} \longrightarrow \mathbb{R}$; $b \longmapsto b^2+1$. $n=2$, $a_0=1$, $a_1=0$, $a_2=1$

We know from the previous example that f is a function. It is interesting to understand Range(f). We claim that $\boxed{\text{Range}(f) = [1,\infty)}$ (∞: Infinity)

where $[1,\infty) = \{x \in \mathbb{R} \mid 1 \leq x\}$. (H.W.)

a) To show Range$(f) \subseteq [1,\infty)$

For every $x \in \mathbb{R}$ (be arbitrary), $x^2 \in \mathbb{R}$ &

$x^2 \geq 0$. Then, $x^2+1 \in \mathbb{R}$ &

$x^2+1 \geq 1$. Hence, $f(x) \in [1,\infty)$.

Hence, Range$(f) \subseteq [1,\infty)$.

b) To show $[1,\infty) \subseteq$ Range(f).

Let $y \in [1,\infty)$ be arbitrary.

We want to show that

there exists $x \in \mathbb{R}$ such that $f(x) = y$. In other words, $x^2 = y - 1$. As $y \in [1, \infty)$, $y - 1 \geq 0$.

On taking square root on both sides, we get

$$x = \sqrt{y-1}.$$

fact: for every $x \in \mathbb{R}, x \geq 0$, $\sqrt{x} \in \mathbb{R}$.

By this fact, we get that $\sqrt{y-1} \in \mathbb{R}$. Hence $f(\sqrt{y-1}) = y$ & $y \in \text{Range}(f)$.

8. Piece-wise function: Let $f: \mathbb{Z} \to \mathbb{Z}$ be defined as

$$f(x) = \begin{cases} x-1 & \text{if } x \text{ is odd} \\ x+1 & \text{if } x \text{ is even} \end{cases}$$

Verify that f is a function. (HW)

We claim that Range$(f) = \mathbb{Z}$. (HW)

By definition, for every $x \in \mathbb{Z}$, $f(x) \in \mathbb{Z}$. Hence, Range$(f) \subseteq \mathbb{Z}$.

Now, let $y \in \mathbb{Z}$ be an arbitrary element. There are two possible cases:

a) y is odd: Then, $y = (y-1)+1$

As y is odd, $y-1$ is even.

Then, by the definition of F, $f(y+1) = y$ & hence $y \in$ Range(f).

b) y is even: Then, $y+1$ is odd & $y = (y+1)-1$. Again, by the definition of f, $f(y+1) = y$.

& $y \in \text{Range}(f)$.

Hence, $\mathbb{Z} \subseteq \text{Range}(f)$.

9) Let $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$; $(m,n) \mapsto m+n$
(m,n)

verify that f is a function (HW)

Also, prove that $\text{Range}(f) = \mathbb{Z}$ (Homework)

4.4.1 Composition of functions:

Definition [function composition]: Let A, B, C, D be non-empty sets, and $f: A \to B$, $G: C \to D$ be functions

with Range(F) $\subseteq$ C. Then, the composition of f and G, denoted G∘f, is the function G∘f: A → D defined as follows: for every $x \in A$, $G \circ f(x) := G(f(x))$ ← $f(x) \in$ Range(F)

uses the fact that Range(F) $\subseteq$ Domain(G).

H.w. What is f∘G? Is F∘G = G∘f for every F, G?

Examples:

1) Let $f: \mathbb{R} \to \mathbb{R}$ and $G: \mathbb{R} \to \mathbb{R}$ such that for every $x \in \mathbb{R}$, $f(x) := x^2$ and $G(x) := x+1$. $G(f(x)) = G(x^2)$, $G(x^2) = x^2+1$

Then, $G \circ f(x) = G((x)^2) = x^2+1$

and $f \circ G(x) = f(x+1) = (x+1)^2$.

This example shows $G \circ f \neq f \circ G$

Equality of functions

2) Let $f: \mathbb{R} \to \mathbb{R}$ and $G: \mathbb{R} \to \mathbb{R}$ be defined as follows: for every $x \in \mathbb{R}$,

$$f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ x-2 & \text{if } x < 0 \end{cases}, \quad G(x) = \begin{cases} x+3 & \text{if } x \geq 4 \\ 2x & \text{if } x < 4 \end{cases}$$

Then, $G \circ f(x) = \begin{cases} x^2+3 & x \geq 2 \\ 2(x^2) & 0 \leq x < 2 \\ 2(x-2) & x < 0 \end{cases}$

& $f \circ G(x) = \begin{cases} (x+3)^2 & x \geq 4 \\ (2x)^2 & 0 \leq x < 4 \\ 2x-2 & x < 0 \end{cases}$

4.4.2 Injective, surjective, and bijective functions

Definition [Injective, surjective, bijective functions]: Let X, Y be non-empty sets and $F: X \to Y$ be a function. Then,

a) F is said to be <u>injective</u> or <u>one-to-one</u> if for every $x_1, x_2 \in X$ if $F(x_1) = F(x_2)$ then $x_1 = x_2$.

Equivalently, if $x_1 \neq x_2$ then $F(x_1) \neq F(x_2)$.

b) F is said to be <u>surjective</u> or <u>onto</u> if for every $y \in Y$, there exists an $x \in X$ such that $F(x) = y$.

Prove that in this case $\text{Range}(F) = Y$.

c) F is said to be bijective if F is both injective and surjective.

Examples:

1) Let $F: \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = x^2 + 1$ for every $x \in \mathbb{R}$.

a) Injectivity: To show that ⌐ Not injective.

if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$. $x_1 = 2$, $x_2 = -2$

$f(x_1) = 5$, $f(x_2) = 5$

b) Surjectivity: Not surjective.

For every $y \in \mathbb{R}$, $\exists\, x \in \mathbb{R}$ s.t. $F(x) = y$. Take $y = -3$.

Suppose $f(x) = y$. $x^2 + 1 = -3$, $x^2 = -4$.

$x \notin \mathbb{R}$ $x = \sqrt{-4}$ or $x = -\sqrt{-4}$

2) Let $a, b \in \mathbb{R}$, $a \neq 0$ & $F: \mathbb{R} \to \mathbb{R}$ be a function such that

for every $x \in \mathbb{R}$, $f(x) := ax+b$.

a) Injectivity:

b) Surjectivity:

3) Let $f: \mathbb{R}^2 \to \mathbb{R}^2$ be a function where $\mathbb{R}^2 := \mathbb{R} \times \mathbb{R}$. Then, for every $(x,y) \in \mathbb{R}^2$, $f(x,y) := (-y,x)$.

a) Injectivity:

b) Surjectivity

Observation 4.4.2.1: Let X, Y be non-empty sets & $f: X \to Y$ be a function. Then, f is surjective if and only if $\text{Range}(f) = Y$.

Homework: Prove this observation.

Lemma 4.4.2.2: Let X, Y, Z be non-empty sets and $f: X \to Y$, $G: Y \to Z$ be functions.

a) If f and G both are injective, then $G \circ f$ is also injective.

b) If f and G are both surjective then $G \circ f$ is also surjective.

c) If f and G are both bijective functions then $G \circ f$ is also bijective.

d) If $G \circ f$ is injective, then f is injective, but G need not be.

e) If $G \circ f$ is surjective, then G is surjective, but G need not be.

Homework : Prove Lemma 4.2.2.1

8th October Lecture-15

Recap: i) Sets, operations on sets & their properties

2) Sequences & equality of sequences

3) functions: Domain, Co-Domain & Range
   - Examples
   - Equality of functions
   - Composition of functions
   - Injectivity, surjectivity and bijectivity.

Lemma 4.4.2.2: let X, Y, Z be non-empty sets and $f: X \to Y, G: Y \to Z$

be functions. Then,

a) If f & G are both injective then $G \circ f$ is injective

b) If f and G are both surjective then $G \circ f$ is also surjective. (HW)

c) If f & G are both bijective then $G \circ f$ is also bijective.

d) If $G \circ f$ is injective then f is injective, but G need not be.

e) If $G \circ f$ is surjective then G is surjective, but f need not be. (HW)

Proof a) Let $x_1, x_2 \in X$ be arbitrary such that $x_1 \neq x_2$. Then, we want to show that $G \circ f(x_1) \neq G \circ f(x_2)$.

We know f is injective. Hence, $f(x_1) \neq f(x_2)$. As $f(x_1), f(x_2) \in Y$ and G is also injective, we get $G(f(x_1)) \neq G(f(x_2))$. Hence, $G \circ f$ is is injective.

b) Let $z \in Z$ be arbitrary. We want to show that there exists an $x \in X$ s.t. $G \circ f(x) = z$.

We know G is surjective. Then, there exists a $y \in Y$ such that $G(y) = z$. As f is also surjective there exists an $x \in X$ such that $f(x) = y$. This implies

$$G(f(x)) = G(y) = z.$$

Hence, $G \circ f$ is surjective.

c) follows from a) & b).

d) Given $G \circ f$ is injective & we want to show that f is also injective. Let $x_1, x_2 \in X$ be arbitary suppose $f(x_1) = f(x_2)$. Then, as G is a function, for every $x \in X$, there exists a unique $z \in Z$ such that $G(f(x)) = y$. This implies

$$G(f(x_1)) = G(f(x_2)).$$

As $G \circ f$ is injective, we get that $x_1 = x_2$. As $f(x_1) = f(x_2)$, we get that f is injective.

Now we show that G need not be injective. Let $X = \{1\}$, $Y = \{a, b\}$ & $Z = \{\alpha\}$. Let $f = \{(1, a)\}$ & $G = \{(a, \alpha), (b, \alpha)\}$

Then, f & $G \circ f$ are injective but G is not.

e) Homework.

Definition [Permutation]: Let X be a non-empty finite set. A bijective function $f: X \to X$ is called a permutation.

For example, let $X = \{1, 2\}$ & $f_1 = \{(1,1), (2,2)\}$ and $f_2 = \{(1,1), (2,1)\}$.

Then, f_1 is a permutation on X, but f_2 is not.

Claim 4.4.2.3: Let X be a non-empty set containing n elements. Then,

there are $n!$ permutations on X, where $n! := n \cdot (n-1) \cdots 2 \cdot 1$

Will prove this later.

4.5 Pigeonhole Principle

Pigeonhole principle is one of the most fundamental tools in counting. In this section, we will see some equivalent formulations of the pigeonhole principle.

Theorem 4.5.1 [Pigeonhole principle]: Let X and Y be two non-empty <u>finite sets</u> having the same number of elements. Let $F: X \to Y$ be a function. Then, f is injective if and only if f is surjective if and only if F is

bijective.

Proof: We know that X and Y have the same number of elements, say n. Let $X = \{x_1, \cdots, x_n\}$ and $Y = \{y_1, \cdots, y_n\}$.

Note that if we prove that f is injective if and only if f is surjective then we get the proof of the entire theorem.

a) Suppose f is surjective. Assume for the contradiction that f is not injective. Then, there exist $x_i, x_j \in X$, $x_i \neq x_j$ such that

$$f(x_i) = f(x_j) = y_k,$$

for some $k \in \{1, \cdots, n\}$. Let $X' := X - \{x_i, x_j\}$ and $Y' := Y - \{y_k\}$. Then, X' has $n-2$

elements and Y' has $n-1$ elements. Let $A := \{f(x) : x \in X'\}$. Then, A has at most $n-2$ elements as X' has $n-2$ elements. Here, we use the fact that f is a function & $f(x)$ is unique for every $x \in X'$. Note that $A \cup \{y_k\} = \text{Range}(f)$, and as A has at most $n-2$ elements we get that $\text{Range}(f)$ has at most $n-1$ elements. We know that f is surjective if and only if $\text{Range}(f) = \text{co-domain}(f)$. Then, as $\text{Range}(f)$ has less than n elements, we get that f is not surjective. This is a contradiction. Hence, f is injective.

b) Suppose f is injective. Then, for every $x_1, x_2 \in X$, $x_1 \neq x_2$, $f(x_1) \neq f(x_2)$. This implies $\{f(x_1), \dots, f(x_n)\}$ = Range (f) contains n distinct elements. Since Y has exactly n elements, it follows immediately that for every $y_i \in Y$, there exists an $x_j \in X$ such that $f(x_j) = y_i$. Hence, f is surjective. ∎

Homework

Lemma 4.5.2: Let X and Y be two non-empty finite sets. Then, X and Y have the same number of elements if and only if there exists a bijective function $f: X \to Y$.

Proof: First we assume that X and Y have the same number of

elements, i.e., $X = \{x_1, \cdots, x_n\}$ and $Y = \{y_1, \cdots, y_n\}$. Let $f : X \to Y$ be the function defined as

$$f = \{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)\}$$

Clearly, f is bijective.

Now we prove the converse. Suppose there is a bijective function $F : X \to Y$. Let $X = \{x_1, \cdots, x_n\}$ and $Y = \{y_1, \cdots, y_m\}$. We want to prove that $n = m$, i.e. $n \leq m$ and $n \geq m$.

As f is bijective, it is both injective and surjective. As f is injective, $f(x_1), \cdots, f(x_n)$ are pairwise distinct elements of Y. This implies $n \leq m$.

As f is surjective, there exist

$x_1, \cdots, x_m \in X$ such that $f(x_1) = y_1, \cdots, f(x_m) = y_m$. We know that f is a function and $(x_1, y_1), \cdots, (x_m, y_m) \in F$. Then, by the definition of a function, we get that $x_1, \cdots, x_m$ should be pairwise distinct elements. Hence $n \geq m$.

Thus, we get $n = m$. ∎

We now prove an important consequence of Theorem 4.5.1 using Lemma 4.5.2. It is often taken as the Pigeonhole principle in various forms.

Theorem 4.5.3 [Pigeonhole principle-version 2]: Let X, Y be non-empty

finite sets such that X has more elements than Y. Let $F: X \to Y$ be an arbitrary function. Then, f is not injective.

Proof: Let $X = \{x_1, \cdots, x_n\}$ & $Y = \{y_1, \cdots, y_m\}$

It is given that $m < n$. It follows from Lemma 4.5.2 that for every function $f: X \to Y$ is not bijective. Further, Theorem 4.5.1 implies that f is not injective. ∎

There are several "fancy" ways to state the above theorem. one of the most popular ways, due to which it is called the

pigeonhole principle, is as follows: If there are more pigeons than the holes they occupy, then there is hole that contains more than one pigeon. Here X & Y are sets of pigeons & holes & $F: X \to Y$ is a function that assigns holes to pigeons. Then, by the above theorem, F is not injective.

- Pigeonhole principle is extensively used in counting. Lets see some examples.

Examples:

a) Prove: In a group of 8 people, there exist two people who were born on the same day of the week.

Let X be the set of 8 people, Y be the set of seven days in a week & $f: X \to Y$ be a function that assigns every person in X to his/her birth day. As X has more elements than Y, by the pigeonhole principle (Theorem 4.5.3) we

get that f is not injective, which means that there exist at least two people in X who are born on the same day.

2) How many students must be there in a class to guarantee that atleast two students recieve the same score in the end-sem, if graded on the scale of 0 to 100?

Let X be the set of students in the class & $Y = \{0, 1, \cdots, 100\}$

Let $f : X \to Y$ be a function that maps every student to their

scores. Then, by the pigeonhole principle (Theorem 4.5.3), if X has at least 10 2 students then f can't be injective & there will always exist atleast two students who score the same marks.

3) A drawer in a dark room contains red, green & blue socks. How many socks must be withdrawn from the drawer to be sure that we have a matching pair?
Answer: 4. Why?

4.6 Equivalence relations & partitions

Definition [Binary relation]: Let A and B be sets. A (binary) relation R from A to B is a subset of $A \times B$. A (binary) relation on A is a subset of $A \times A$.

- When R is a relation from A to B, then elements $a \in A$ & $b \in B$ are said to be <u>R-related</u> if and only if $(a, b) \in R$. It is denoted by <u>$a R b$</u>.

Definition [Domain of a relation]:

Let R be a relation from a set A to a set B, i.e., $R \subseteq A \times B$. Then, the domain of R, denoted Domain(R) is defined as

$$\text{Domain}(R) := \{a \in A : \exists\, b \in B \text{ s.t. } (a,b) \in R\}.$$

Examples:

1) Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$

Let $R = \{(1,a), (1,b), (3,c)\}$. Then, R is a relation but not a function.

- It is easy to verify that every function is a relation, but

→ Homework

every relation is not a function.

2) Let $R = \{(x,y) \in \mathbb{R}\times\mathbb{R} : x < y\}$.
Then, it is not difficult to verify that Domain$(R) = \mathbb{R}$. H.W.

Definition [Reflexive, Symmetric, Transitive & Equivalence relation]:
Let A be a set & R be a relation on A. ($R \subseteq A\times A$) Then,

a) R is said to be reflexive if for every $a \in A$, $(a,a) \in R$.

b) R is said to be symmetric if for every $(a,b) \in R$, $(b,a) \in R$.

c) R is said to be transitive if for every $(a,b),(b,c) \in R$, we have $(a,c) \in R$.

d) R is called an equivalence relation if it is symmetric, reflexive & transitive.

Examples:

1) Let $X=\{1,2,3\}$ & $Y=\{a,b\}$ & $R \subseteq X \times Y$ be arbitrary. Then, R is neither reflexive, nor symmetric, nor transitive.

2) Let $X=\{1,2,3\}$ & $R=\{(1,1),(2,2),(3,3),(1,2)\}$. Then, R is reflexive, transitive, but

not symmetric as $(1,2) \in R$ but $(2,1) \notin R$. ✗

3) Let $X = \{1,2,3\}$ and $R = \{(1,2), (2,1)\}$

Then, R is symmetric but neither reflexive nor transitive

3') Let $R = \phi$.

4) Let X be a non-empty set & $R = \phi$.

- As X is non-empty, R is not reflexive.
- R is symmetric: It requires if $(a,b) \in R$ then $(b,a) \in R$. As $(a,b) \in R$ is false, we get from the truth table of $\Rightarrow$ that the

implication stated above is true.

- R is transitive: Can be argued the same as the last point.

5. Let X be a non-empty set & $R = \{(x,y) \in X \times X : x = y\}$.

Then, R is an equivalence relation.

6. Let $\leq$ be the less than or equal to relation on $\mathbb{Z}$.

$\leq := \{(x,y) \in \mathbb{Z} \times \mathbb{Z} : x \leq y\}$.

Then, $\leq$ is reflexive & transitive, but not symmetric.

7. Let $\mathbb{Z}^* := \mathbb{Z} \setminus \{0\}$. Let R

be a relation on $\mathbb{Z} \times \mathbb{Z}^*$ defined as follows: for every $a, c \in \mathbb{Z}$ & $b, d \in \mathbb{Z}^*$

$((a,b),(c,d)) \in R$ if and only if

$$ad = bc.$$

a) R is reflexive: ✓

b) R is symmetric: Let $((a,b),(c,d)) \in R$. This implies

$$ad = bc,$$

which is the same as

$$cb = da.$$

By definition of R, it happens if and only if $((c,d),(a,b)) \in R$.

Hence R is symmetric.

c) R is transitive: Let $((a,b),(c,d)) \in R$, $((c,d),(e,f)) \in R$ & we want to show that $((a,b),(e,f)) \in R$.

This means we are given

$$ad = bc \text{ & } cf = de,$$

which can also be written as

$$c = \frac{de}{f} \quad (\text{over } \otimes).$$

This equation is valid as $f \in \mathbb{Z}^*$. On substituting this value of c in $ad = bc$, we get

$$ad = \frac{bde}{f},$$

which implies

$$adf = bde.$$

$\Rightarrow d(af - be) = 0$ [using commutativity & additive inverse].

Thus, by the no non-zero divisors, either $d = 0$ or $af - be = 0$.

As $d \in \mathbb{Z}^*$, $d \neq 0$. Hence,

$$af - be = 0.$$

This implies $((a,b),(e,f)) \in R$.

Hence, R is transitive.

d) R is an equivalence relation

Remark: R is used to construct

rational numbers using integers.

13th October Lecture-16

Recap:

1) Sets, operation on sets
2) Sequences
3) functions, composition of functions, injective, surjective & bijective
4) permutation function
5) Pigeon hole principle
6) Binary relations.